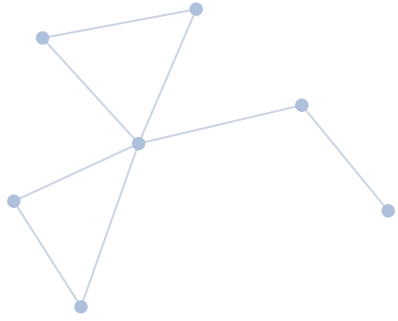




HACKATHON SUBMISSION

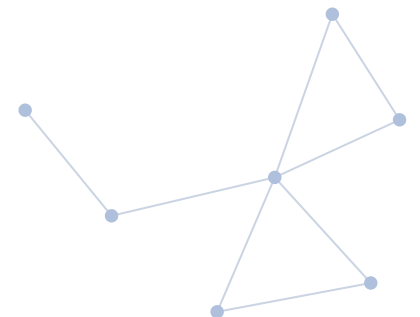
Proof-Of-Work

Decentralized Labor Verification & AI-Powered Trust Infrastructure

Team “Peanut Butter”

[Bharanidharan S] · [Suryanarayanan S] · [Kirubalan S]

[Build2Gether AI Hacathon] · 2026



The Trust Gap in Global Labor

Millions of workers contribute valuable skills but lack a reliable way to prove their professional history.

CURRENT PROBLEMS

- Worker records are controlled by centralized platforms
- Informal workers struggle to verify experience
- Fake credentials reduce trust
- Supply chains lack labor transparency

THE RESULTING FLOW



Introducing Proof-Of-Work

A decentralized verification ecosystem where workers own their professional identity and organizations can verify authentic contributions.



Limitations of Current Approaches

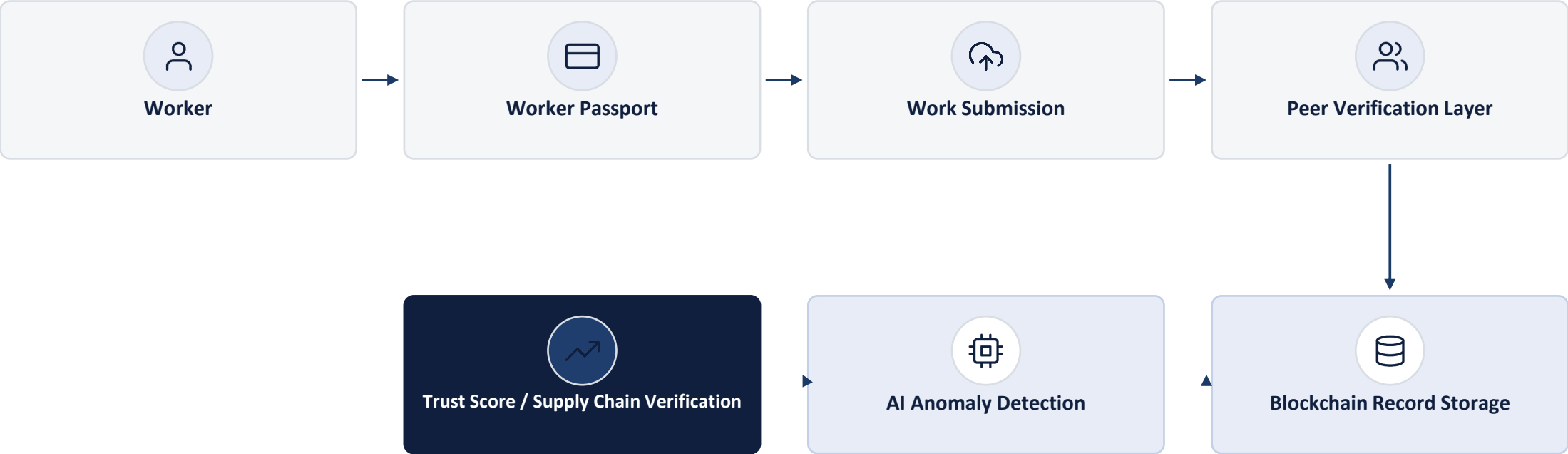
TRADITIONAL SYSTEM

- ✗ Centralized databases
- ✗ Organization controlled
- ✗ Difficult portability
- ✗ Limited transparency

OUR APPROACH

- ✓ Worker-owned identity
- ✓ Distributed verification
- ✓ Portable reputation
- ✓ Transparent records

How The System Works



Core Features



Worker Passport

Digital identity for professional history.



Peer Verification

Community-driven validation of contributions.



Blockchain Records

Tamper-resistant verification history.



AI Fraud Detection

Identifies suspicious verification patterns.



Supply Chain Transparency

Connects products with verified labor.

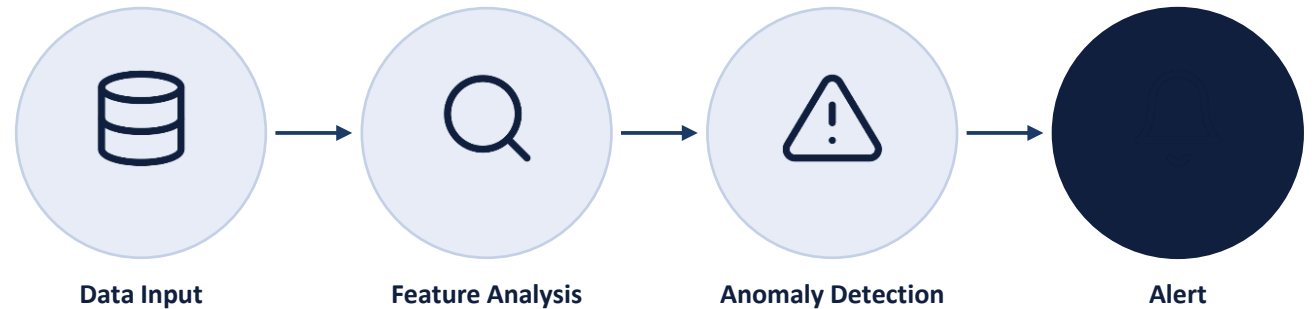
Intelligent Trust Monitoring

The AI layer analyzes verification behavior to detect anomalies.

DETECTION EXAMPLES

- Suspicious validator patterns
- Fake contribution claims
- Impossible working hours
- Location inconsistencies
- Abnormal activity frequency

ML PIPELINE



Anomalies are scored and routed for review before they affect a worker's trust profile.

Technology Stack



Frontend

HTML
CSS
JavaScript



Backend

Application Logic



Blockchain

Smart Contract Concepts
Decentralized Identity



AI

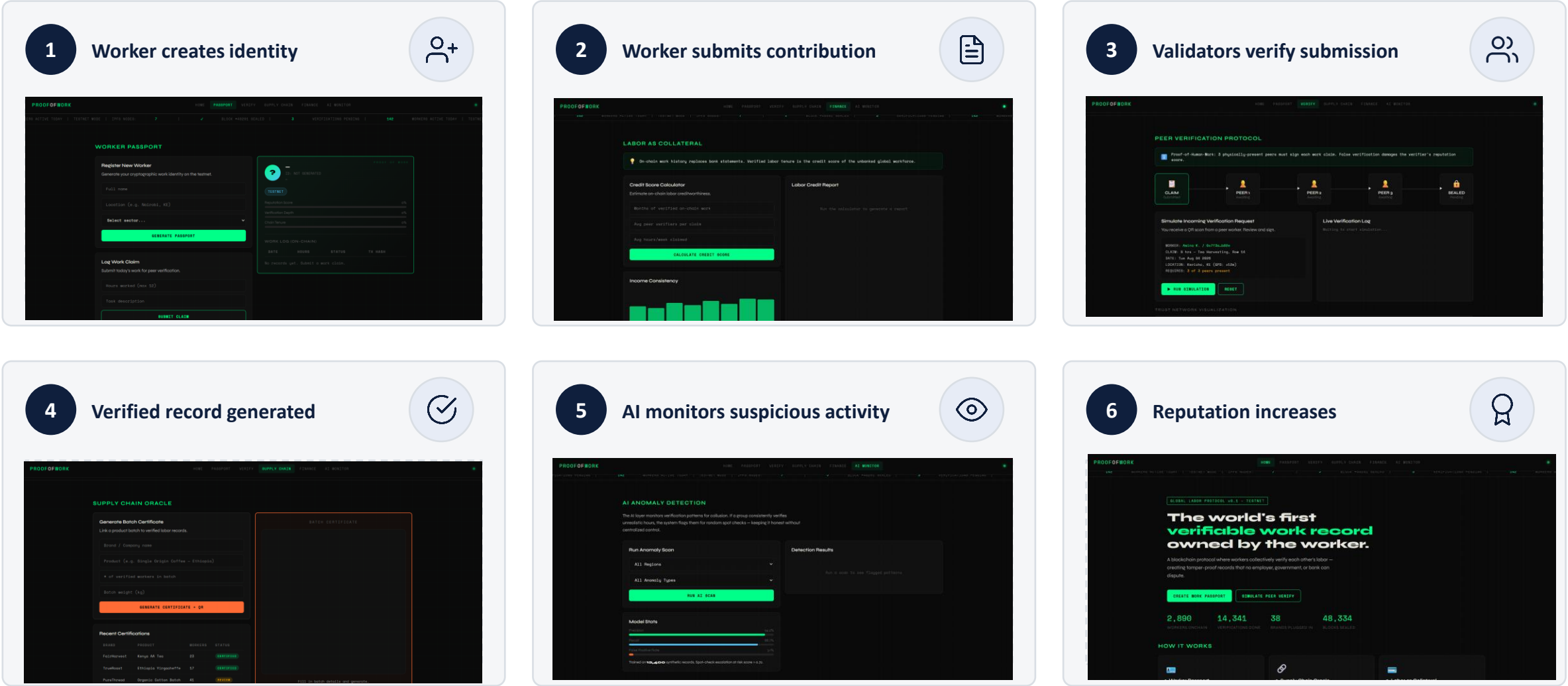
Machine Learning
Anomaly Detection



Infrastructure

Docker
IPFS (future integration)

Application Flow



Real-World Applications



Workers

- Own professional identity
- Build portable reputation



Organizations

- Verify workforce authenticity
- Improve accountability



Industries

- Ethical supply chains
- Gig economy
- Global workforce platforms

Future Improvements





Building Trust For The Future Of Work

“Proof-Of-Work transforms labor history from a centralized record into a worker-owned digital asset.”

Thank You

Team “Peanut Butter”

[Bharanidharan S] · [Suryanarayanan S] · [Kirubalan S]

